

Database Architecture & ERD Design Report (Phase 1)

Sports Event Ticket Booking & Reservation Platform

Technical Documentation | Relational Database Systems

Database Engine:	PostgreSQL 16+ (UTF-8)	Normalization:	Third Normal Form (3NF)
Platform Type:	Sports Event Ticketing	Architecture:	Domain-Driven Partitioning

1. Introduction & Architectural Objectives

The primary objective of Phase 1 is to conduct a detailed domain analysis for a high-traffic sports event ticket booking system and deliver an optimal logical schema free from data redundancies. The relational database schema strictly adheres to the **Third Normal Form (3NF)** while laying the groundwork for high-performance query execution ($O(\log N)$) under high concurrent load.

2. Database Entities & Schema Overview

The system architecture consists of **8 interconnected, domain-partitioned tables**:

- **users Table:** Manages user profile demographics, password hashes, role-based privileges (audience vs. support), and soft-deletion flags (`is_active`).
- **tickets Table:** Serves as the generalized base table holding shared match attributes including match title, sport type, organizer, venue, city, match schedule, price, and remaining capacity.
- **Extension Tables** (`football_details`, `volleyball_details`, `basketball_details`): Specialized 1-to-1 extension tables storing sport-specific metadata (such as VAR availability, parking, indoor/outdoor features, section, row, and seat allocation) without polluting the base ticket table with excessive NULL values.
- **reservations Table:** Handles temporary reservation state lifecycles with a strict 10-minute hold timeout prior to ticket purchase confirmation.
- **payments Table:** Captures financial transactions linked via 1-to-1 relations with reservation records.
- **reports Table:** Logs customer support disputes, complaints, and ticket issues submitted by spectators for administrative review.

3. Normalization Analysis & 3NF Justification

All tables strictly satisfy the criteria for Third Normal Form (3NF):

1. **First Normal Form (1NF):** All attributes are atomic. Multi-valued attributes and repeating groups are eliminated.
2. **Second Normal Form (2NF):** The database is in 1NF and every non-prime attribute is fully functionally dependent on the primary key (`PRIMARY KEY`), with no partial dependencies.
3. **Third Normal Form (3NF):** The schema contains no transitive functional dependencies. Sport-specific attributes are partitioned into dedicated 1-to-1 child tables, preventing non-key attributes from depending on other non-key attributes and eliminating redundant NULL fields.

Table 1: Schema Architecture Summary: Keys, Relations, and 3NF Rationale

Table Name	Primary Key	Foreign Key(s)	3NF Design Rationale
users	user_id	-	Unique email/phone; decoupled authentication state.
tickets	ticket_id	-	Generalized match meta-data shared across all sports.
football_details	ticket_id	ticket_id → tickets	Isolates outdoor pitch & VAR details.
volleyball_details	ticket_id	ticket_id → tickets	Isolates indoor hall & seating section details.
basketball_details	ticket_id	ticket_id → tickets	Isolates court-side access & league features.
reservations	reservation_id	user_id, ticket_id	Manages transient hold lifecycles & expiration logic.
payments	payment_id	reservation_id, user_id	1-to-1 financial ledger decoupled from reservations.
reports	report_id	user_id, reservation_id	Independent complaint logging; avoids transitive dependencies.

4. Integrity Constraints & Domain Business Logic

To guarantee data consistency at the engine layer, declarative constraints are defined:

- **Value Constraints (CHECK):** Non-negative pricing ($\text{price} \geq 0$), non-negative seating capacity ($\text{remaining_capacity} \geq 0$), and valid transaction amounts ($\text{amount} \geq 0$).
- **Temporal Logic Constraints:** Valid reservation time windows ($\text{reserved_at} < \text{expires_at}$).
- **Domain Enumerations (ENUM):** Restricted domain types for user roles (`user_role`), reservation lifecycle status (`reservation_status`), payment states (`payment_status`), and sport categories (`sport_type_enum`).
- **Referential Actions (ON DELETE):** Utilizes CASCADE for cascading cleanups on user removal and SET NULL to preserve historical support reports.

5. Performance Indexing & Query Optimization

To accelerate high-frequency queries and wildcard text lookups:

- **Compound B-Tree Indexes:** Compound B-Tree index on (`sport_type`, `city`) and chronological index on `match_date` ASC for rapid filtering and sorting.
- **GIN Trigram Indexes (pg_trgm):** Advanced Generalized Inverted Index (GIN) trigram indexes on `venue_name` and `title` supporting fast sub-string wildcard searches (ILIKE).

6. Entity-Relationship Data Dictionary & Structural Analysis

The structural breakdown of all system entities, primary attributes, foreign key relationships, and cardinalities is formally detailed in Table 3 below.

Table 2: Data Dictionary: Entities, Attributes, Foreign Keys, and Relationships

Entity / Table	Attributes & Keys	Entity Relationships	Cardinality & Target
users	user_id (PK), full_name, phone_number, email, password_hash, role, city, is_active, created_at	Makes Reservation Submits Report	1:N → reservations 1:N → reports
tickets	ticket_id (PK), title, sport_type, organizer, venue_name, city, match_date, price, remaining_capacity	Booked In Has Football Has Volleyball Has Basketball	1:N → reservations 1:1 → football_details 1:1 → volleyball_details 1:1 → basketball_details
football_details	ticket_id (PK/FK), league_name, stand_section, row_number, seat_number, has_var, parking_access	Extension of Base Ticket	1:1 → tickets
volleyball_details	ticket_id (PK/FK), tournament_name, hall_name, is_indoor, seat_section, row_number, seat_number	Extension of Base Ticket	1:1 → tickets
basketball_details	ticket_id (PK/FK), league_name, hall_name, court_side_access, seat_section, row_number, seat_number	Extension of Base Ticket	1:1 → tickets
reservations	reservation_id (PK), user_id (FK), ticket_id (FK), status, reserved_at, expires_at	Paid By Logged In Report	1:1 → payments 1:N → reports
payments	payment_id (PK), reservation_id (FK/Unique), user_id (FK), amount, status, payment_method, paid_at	Settles Reservation	1:1 → reservations
reports	report_id (PK), user_id (FK), reservation_id (FK), category, report_text, status, created_at	Filed By User	N:1 → users

7. Phase 1 Executive Summary

The first phase of the Sports Ticket Platform project successfully establishes a fully normalized, high-performance 3NF relational database architecture in PostgreSQL. Key milestones achieved in this phase include:

- **Comprehensive Entity Modeling:** Designed 8 interconnected tables encapsulating core user

management, generic ticket metadata, sport-specific extension models, transient reservations, financial ledgers, and support ticket logging.

- **Data Integrity & Business Logic:** Implemented strict declarative check constraints, temporal logic validations, custom PostgreSQL ENUM types, and cascading foreign key actions to ensure engine-level consistency.
- **Advanced Performance Indexing:** Configured B-Tree composite indexes for multi-column filtering and GIN trigram indexes (`pg_trgm`) for high-speed sub-string fuzzy text searching in both Persian and English.
- **Production Readiness:** Verified DDL and DML scripts against analytical scenarios and PL/pgSQL stored procedures, creating a seamless foundation for downstream FastAPI backend integration and Redis caching in Phase 2.

Phase 1 Architectural Conclusion

The designed database schema fulfills all requirements for downstream integration (FastAPI Web Services, SQL Procedure Layer, and Redis Caching), achieving industry-grade data integrity, full 3NF normalization, and query execution efficiency.